Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2000-II-Q05] | Part A (Q1-Q5)

Choose five of eight distinct rings and place them on four labeled fingers of one hand, allowing empty fingers and distinguishing ring order on each finger. Let N count arrangements. Read its first three nonzero decimal digits from the left.

从八个不同戒指中选五个戴在一只手的四根有标识手指上，允许空指，每指戒指顺序不同视为不同。设排法数为 N，求从左起前三个非零十进制数字组成的数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2021-I-Q04] | Part A (Q1-Q5)

Count positive integer triples $a < b < c$ with $a + b + c = 66$.

求满足 $a < b < c$ 且 $a + b + c = 66$ 的正整数三元组个数。

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2003-I-Q09] | Part B (Q6-Q10)

Count four-digit decimal integers whose first two digits sum to the same value as their last two digits.

求前两位数位和等于后两位数位和的四位十进制整数个数。

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2009-I-Q09] | Part B (Q6-Q10)

Three ordered prices are integers from one to 9999, and their concatenated digits consist of four ones and three threes. Count all possible ordered price triples.

三件有序商品价格为一至 9999 的整数，三价格串联后的数字恰为四个一、三个三。求可能的有序价格三元组数。

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2009-I-Q10] | Part B (Q6-Q10)

Fifteen labeled circular seats hold five distinct members of each of groups M, V, E, with seat one occupied by M and seat fifteen by E. Clockwise changes of group are allowed only $M \to V, V \to E, E \to M$. If total arrangements are $N(5!)^3$, find N.

十五个有编号圆桌座位坐三组各五名不同成员，第一座为 M 组，第十五座为 E 组。按顺时针，换组只允许 $M \to V, V \to E, E \to M$。若总安排数为 $N(5!)^3$，求 N。

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2011-II-Q07] | Part B (Q6-Q10)

Arrange five identical green marbles and some identical red marbles in a row, with equally many same-color and different-color adjacent pairs. Maximize the red count m, and let N count the valid rows at that maximum. Find $N \bmod 1000$.

将五颗相同绿珠与若干相同红珠排成一行，要求同色相邻对数与异色相邻对数相等。令最大红珠数为 m，达到此最大值的排列数为 N，求 N 模 1000。

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2025-II-Q10] | Part B (Q6-Q10)

Choose eight occupied chairs among sixteen in a row, with no occupant having occupied neighbors on both sides. If there are N chair subsets, find $N \bmod 1000$.

十六把椅子排成一行，选八把坐人，要求无人左右两侧都有人坐。若合法椅子子集数为 N，求其除以 1000 的余数。

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2008-II-Q12] | Part C (Q11-Q15)

Arrange ten identical blue and nine identical green flags in ordered stacks on two distinguishable poles, each nonempty, with no adjacent greens on a pole. Find the count modulo 1000.

将十面相同蓝旗、九面相同绿旗按次序挂在两根不同旗杆上，每杆非空且绿旗不相邻，求排法数除以一千的余数。

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2018-II-Q15] | Part C (Q11-Q15)

Count integer-valued functions on $\{0,1,\ldots,6\}$ with $f(0)=0, f(6)=12$ and $|x-y| \le |f(x)-f(y)| \le 3|x-y|$ for every pair of inputs.

求定义在 $\{0,1,\ldots,6\}$ 上的整数值函数数，满足 $f(0)=0, f(6)=12$，且任意输入对均有 $|x-y| \le |f(x)-f(y)| \le 3|x-y|$。

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Section 4.3 Compositions, partitions & generating functions

整数拆分、分拆与生成函数

[AIME-2022-II-Q13] | Part C (Q11-Q15)

The expression $(x^{2310}-1)^6/[(x^{105}-1)(x^{70}-1)(x^{42}-1)(x^{30}-1)]$ agrees with an integer-coefficient polynomial $P(x)$ on $0<x<1$. Find the coefficient of x^{2022}.

表达式 $(x^{2310}-1)^6/[(x^{105}-1)(x^{70}-1)(x^{42}-1)(x^{30}-1)]$ 在 $0<x<1$ 上等于某整系数多项式 $P(x)$。求 x^{2022} 的系数。

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